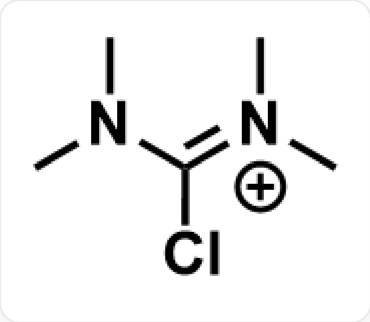
Question

The structure of the Vilsmeier salt cation is shown in the figure,



CN(C)/C(Cl)=[N+](C)/C

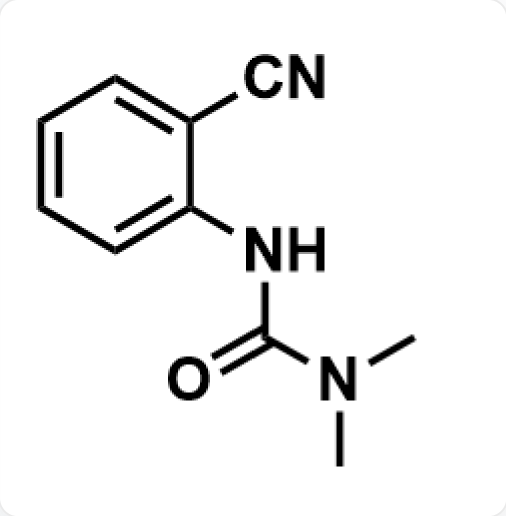
A laboratory attempted to synthesize compound **A** by reacting excess Vilsmeier salt with an equivalent amount of base and o-aminobenzamide, but found that no **A** was formed. Instead, a yellow, low-melting-point, non-crystalline compound **B** was obtained. Liquid chromatography-mass spectrometry (LC-MS) analysis revealed that the molecular weight of **B** was 216.28.

The laboratory then heated compound **B** with a large excess of hydroxylamine hydrochloride and an equivalent amount of sodium acetate in a methanol-water solution, intending to prepare the seven-membered cyclic compound **C** with a molecular weight of 204.23. However, thin-layer chromatography (TLC) development using dichloromethane and petroleum ether containing triethylamine revealed a non-fluorescent compound **D** and a fluorescent compound. LC-MS and ^{1}H NMR analysis confirmed that the fluorescent compound was **A**. LC-MS analysis showed that the molecular weight of **D** was also 204.23.

X-ray diffraction (XRD) confirmed that compound **D** possesses a hexa-hexacyclic system, which was not the intended structure of **C**. To investigate the reaction mechanism, diethylhydroxylamine was used instead of hydroxylamine hydrochloride and sodium acetate, but no analogs of **D** were formed.

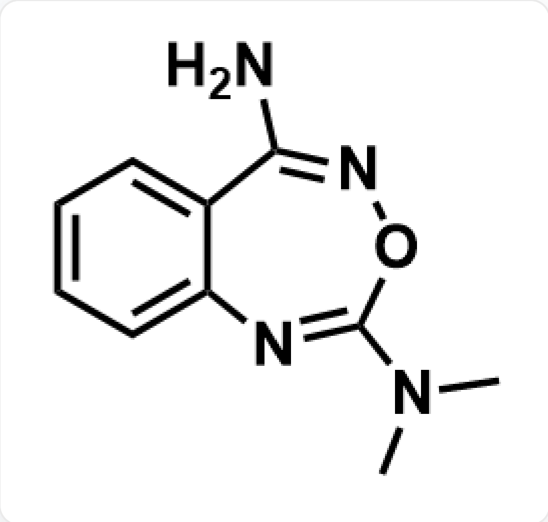
The following options each contain three structural images: the first is the possible structure of **A**, the second is the possible structure of **C**, and the third is the possible structure of **D**. Select the option where all three structures are correct.

A. A:



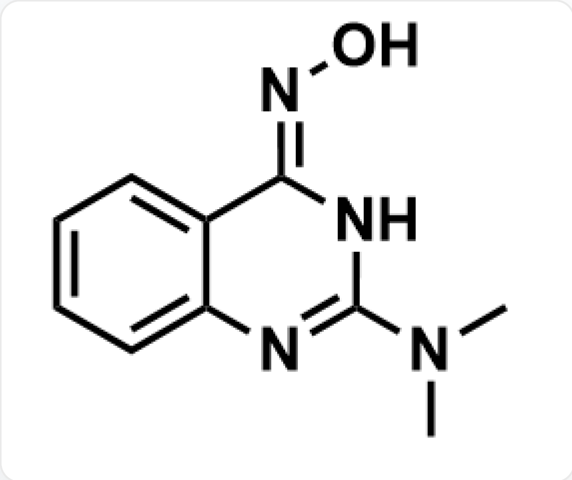
O=C(N(C)C)NC1=CC=CC=C1C#N

C:



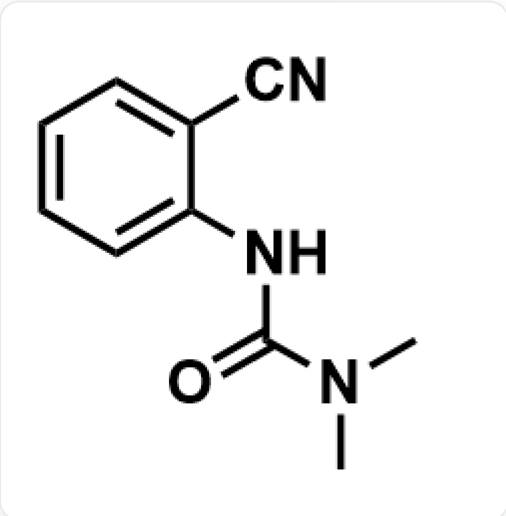
NC1=NOC(N(C)C)=NC2=CC=CC=C21

D:



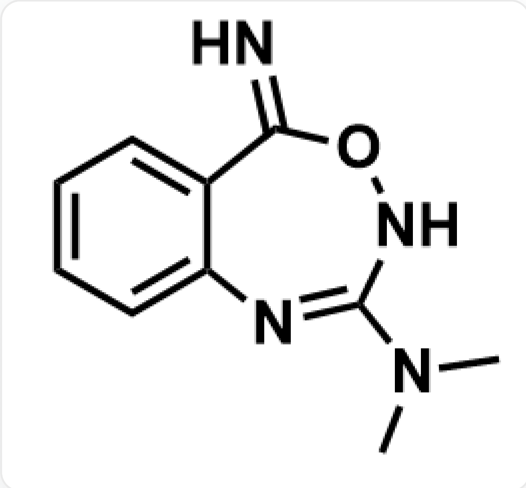
O/N=C1NC(N(C)C)=NC2=CC=CC=C2\1

B. A:



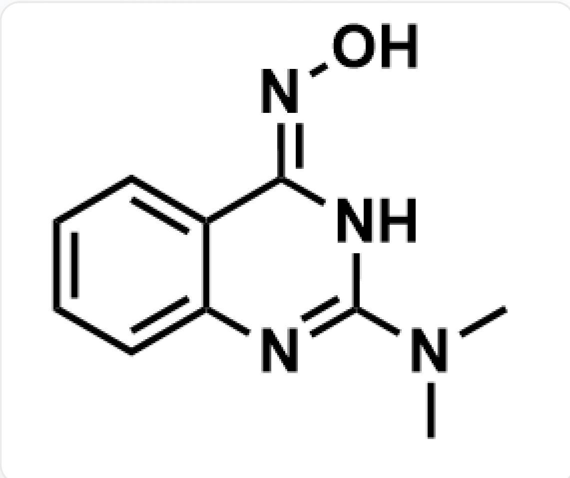
O=C(N(C)C)NC1=CC=CC=C1C#N

C:



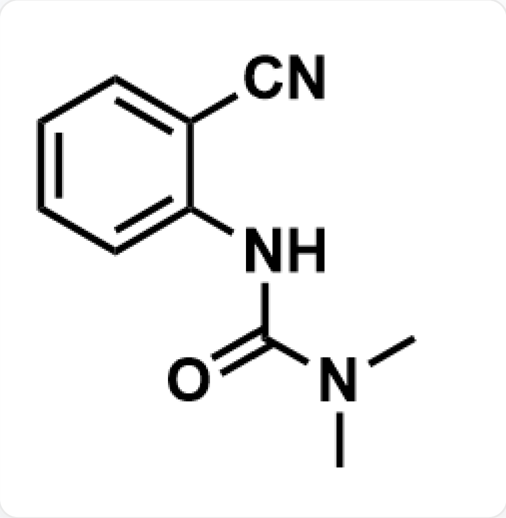
N=C1C2=CC=CC=C2N=C(N(C)C)NO1

D:



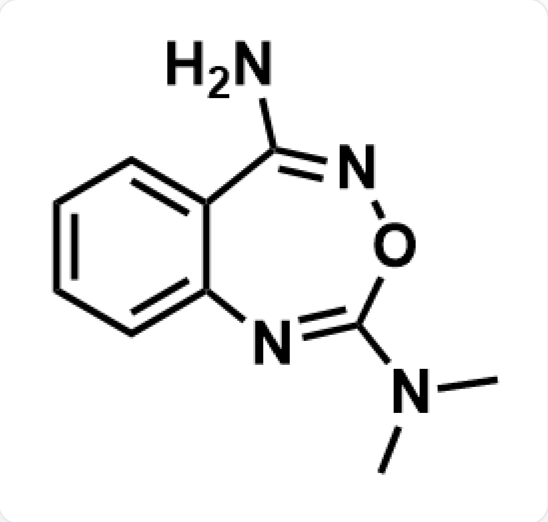
O/N=C1NC(N(C)C)=NC2=CC=CC=C2\1

C. A:



O=C(N(C)C)NC1=CC=CC=C1C#N

C:

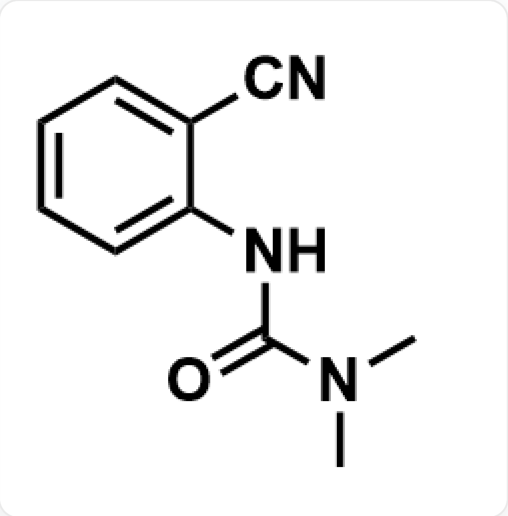


NC1=NOC(N(C)C)=NC2=CC=CC=C21

D:

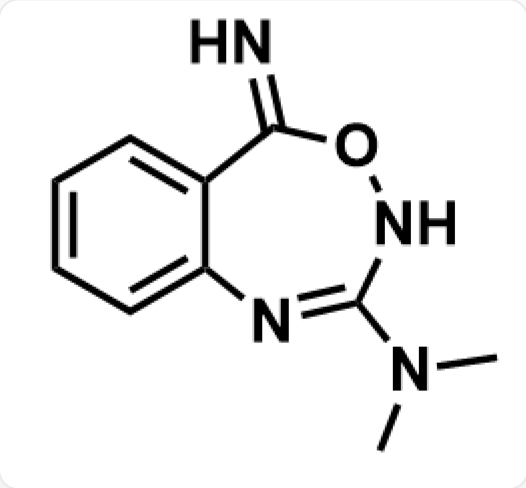
CN(C)C1=NC2=CC=CC=C2C(ON)=N1

D. A:



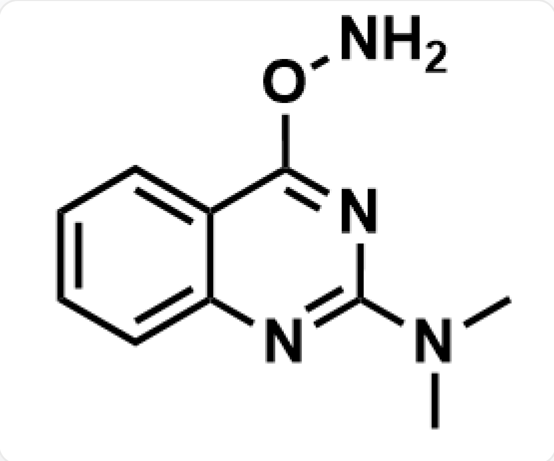
O=C(N(C)C)NC1=CC=CC=C1C#N

C:



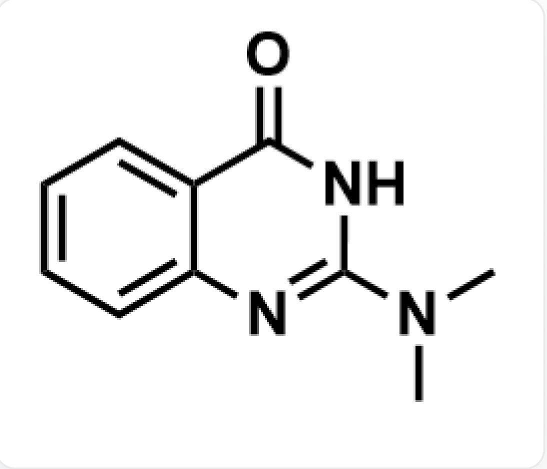
N=C1C2=CC=CC=C2N=C(N(C)C)NO1

D:



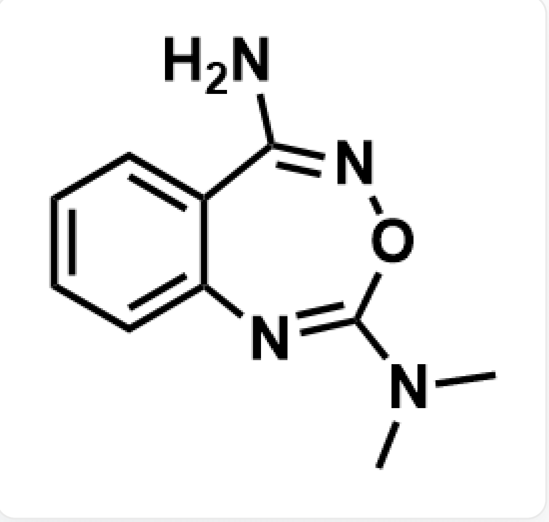
CN(C)C1=NC2=CC=CC=C2C(ON)=N1

E. A:



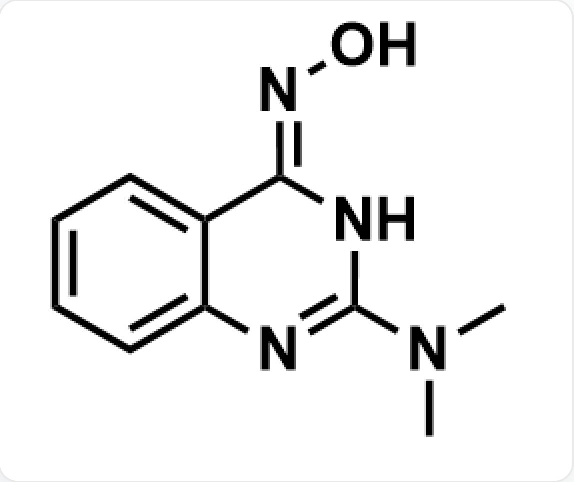
O=C1C2=CC=CC=C2N=C(N(C)C)N1

C:



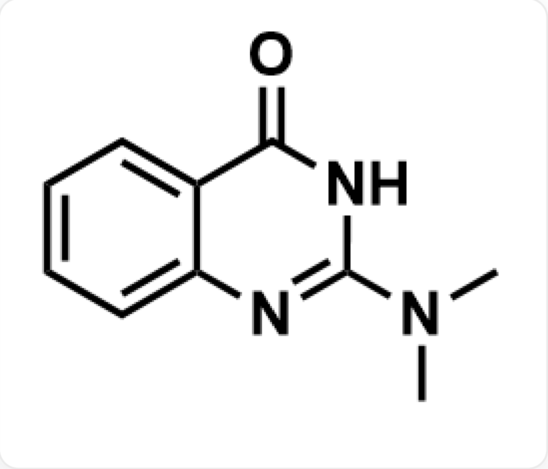
NC1=NOC(N(C)C)=NC2=CC=CC=C21

D:



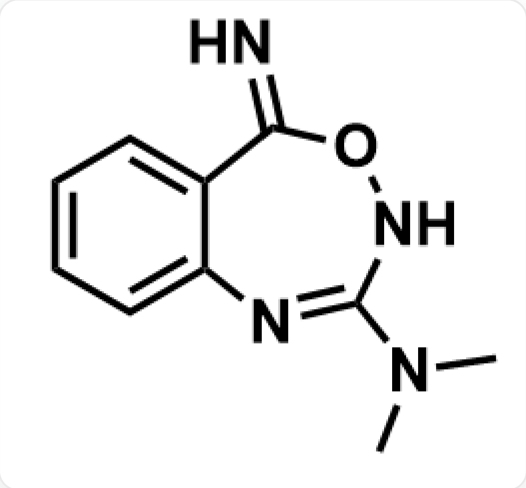
O/N=C1NC(N(C)C)=NC2=CC=CC=C2\1

F. A:



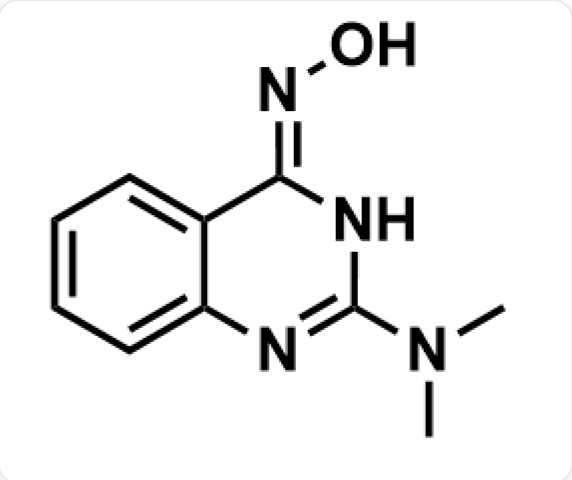
O=C1C2=CC=CC=C2N=C(N(C)C)N1

C:



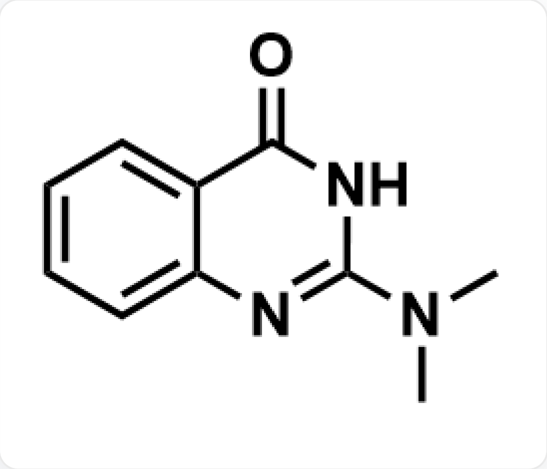
N=C1C2=CC=CC=C2N=C(N(C)C)NO1

D:



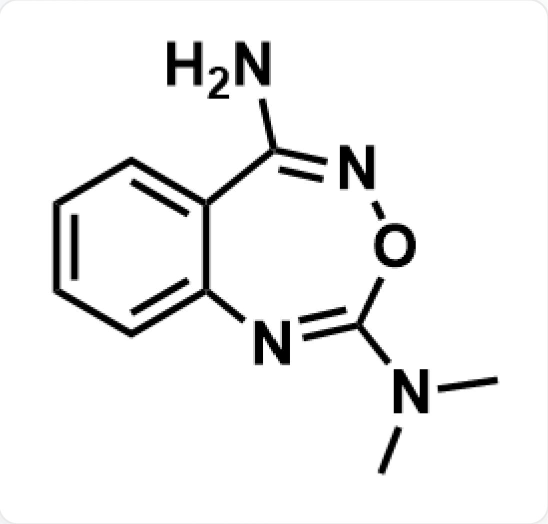
O/N=C1NC(N(C)C)=NC2=CC=CC=C2\1

G. A:



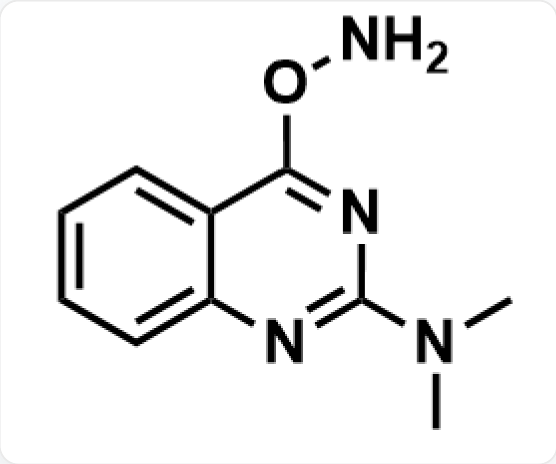
O=C1C2=CC=CC=C2N=C(N(C)C)N1

C:



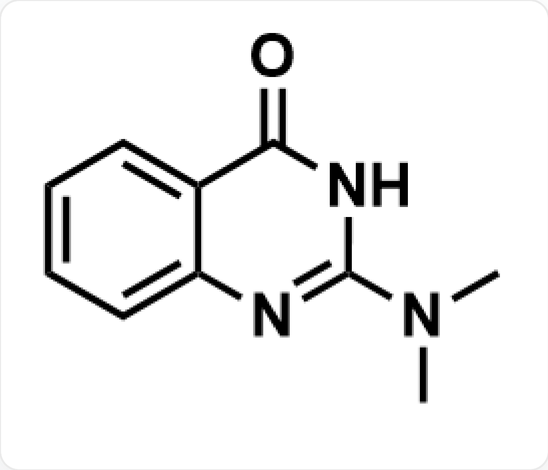
NC1=NOC(N(C)C)=NC2=CC=CC=C21

D:



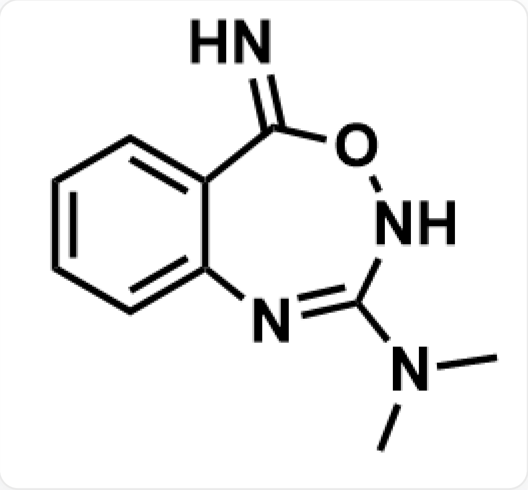
CN(C)C1=NC2=CC=CC=C2C(ON)=N1

H. A:



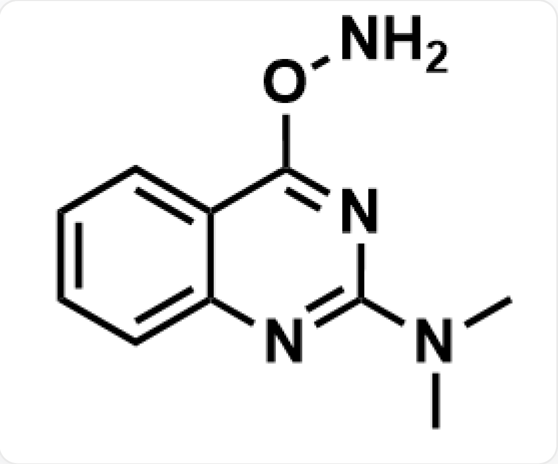
O=C1C2=CC=CC=C2N=C(N(C)C)N1

C:



N=C1C2=CC=CC=C2N=C(N(C)C)NO1

D:



CN(C)C1=NC2=CC=CC=C2C(ON)=N1

I. None of the above options are correct

Answer

Correct Answer: E

Detailed Explanation

(1)

Upon observing the substrate, both the amide oxygen and the amino nitrogen of anthranilamide are nucleophilic sites and can react with the Vilsmeier salt.

CHECKPOINT 1 PTS

Both the amide oxygen and the amino nitrogen of the substrate are nucleophilic sites

The question states that **A** exhibits fluorescence, indicating that **A** should possess a large conjugated system and a relatively rigid skeleton, and cannot consist of only one benzene ring.

It can be inferred that the formation of compound **A** involves the reaction of the amino group with the Vilsmeier salt. During the formation of **A**, the amino group nucleophilically attacks the Vilsmeier salt, followed by an intramolecular attack by the amide to form a six-membered ring. The elimination of dimethylamine yields **A**, which contains a conjugated system.

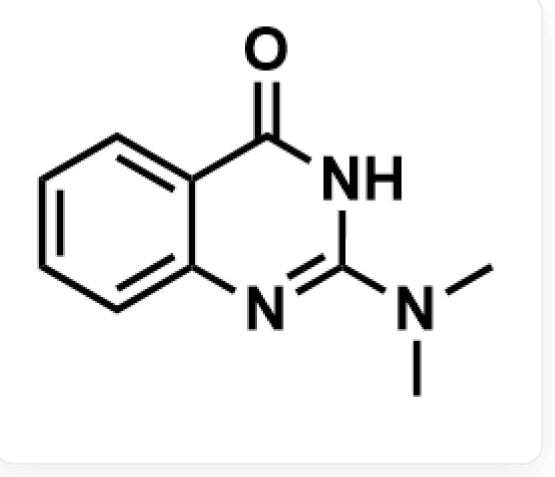
CHECKPOINT 1 PTS

A exhibits fluorescence and cannot consist of only one benzene ring

CHECKPOINT 1 PTS

The formation of **A** involves the reaction of the amino group with the Vilsmeier salt

The structure of compound **A** is shown below:



O=C1C2=CC=CC=C2N=C(N(C)C)N1

CHECKPOINT 2 PTS

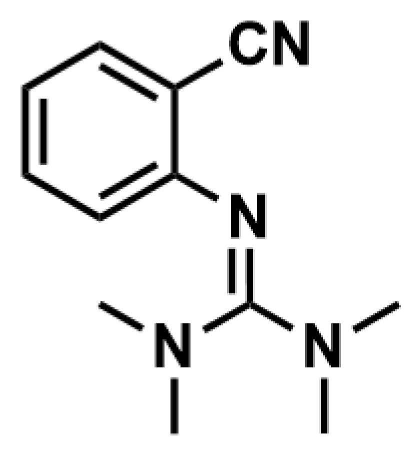
The structure of **A** is O=C1C2=CC=CC=C2N=C(N(C)C)N1

If the amide reacts with the Vilsmeier salt first, the amide oxygen nucleophilically attacks the Vilsmeier salt. The Vilsmeier salt acts as a dehydrating agent, formally eliminating water to yield anthranilonitrile and dimethylaminourea. Based on the molecular formula, dimethylaminourea further reacts with the amino group to eliminate water, resulting in **B**.

CHECKPOINT 1 PTS

The formation of **B** involves the reaction of the amide with the Vilsmeier salt first

The structure of compound **B** is shown below:



CN(C)/C(N(C)C)=N/C1=CC=CC=C1C#N

CHECKPOINT 2 PTS

The structure of **B** is CN(C)/C(N(C)C)=N/C1=CC=CC=C1C#N

(2)

Compound **C** is formed by the reaction of compound **B** with hydroxylamine. Based on the molecular weight, **C** contains one more hydroxylamine and one less dimethylamine than **B**, indicating a substitution reaction. Since **C** contains a seven-membered ring, an intramolecular cyclization also occurs. Considering the reactivity of compound **B**, the most electrophilic site is the cyano group, and hydroxylamine should react with the cyano group first.

CHECKPOINT 1 PTS

Hydroxylamine should react with the cyano group first

Both the nitrogen and oxygen of hydroxylamine are nucleophilic sites, and further analysis requires information about compound **D**.

The question states: Using diethylhydroxylamine as a reactant fails to yield an analog of **D**, thereby excluding the possibility of nucleophilic attack by the oxygen end of hydroxylamine. Thus, the nitrogen end of hydroxylamine is connected to the carbon atom of the six-membered ring.

CHECKPOINT 1 PTS

Diethylhydroxylamine as a reactant fails to yield an analog of D, indicating the nitrogen end of hydroxylamine acts as the nucleophilic site

Therefore, the carbon corresponding to the cyano group in **B** is connected to two nitrogen atoms in **C** and **D**, namely the nitrogen of the cyano group and the nitrogen of hydroxylamine.

CHECKPOINT 1 PTS

The carbon corresponding to the cyano group in **B** is connected to two nitrogen atoms in **C** and **D**

The reaction of the cyano group with hydroxylamine can yield an amidoxime intermediate with the structure N/C(C1=CC=CC=C1/N=C(N(C)C)\N(C)C)=N\O. The guanidine carbon atom in the molecule is an electrophilic site, and either the oxygen atom or the amino nitrogen atom of the amidoxime can act as a nucleophile, forming a seven-membered ring or a six-membered ring, respectively. Eliminating one molecule of dimethylamine yields **C** and **D**, respectively.

CHECKPOINT 1 PTS

The reaction of the cyano group with hydroxylamine yields the amidoxime intermediate N/C(C1=CC=CC=C1/N=C(N(C)C)\N(C)C)=N\O

CHECKPOINT 1 PTS

The oxygen atom or amino nitrogen atom of the amidoxime can nucleophilically attack the guanidine carbon atom to form a seven-membered/six-membered ring

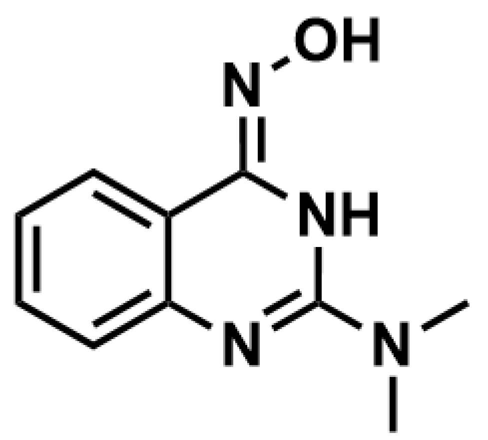
The structure of compound **C** is shown below:

NC1=NOC(N(C)C)=NC2=CC=CC=C21

CHECKPOINT 2 PTS

The structure of **C** is NC1=NOC(N(C)C)=NC2=CC=CC=C21

The structure of compound **D** is shown below:



O/N=C1NC(N(C)C)=NC2=CC=CC=C2\1

CHECKPOINT 2 PTS

The structure of **D** is O/N=C1NC(N(C)C)=NC2=CC=CC=C2\1

In conclusion, option E is correct.